Protocol for the Expert Evaluations

Blind elicitation

Before showing you my work, I want to ask you some questions to see what an ontology for explainable machine learning pipelines applied to a healthcare screening case should cover.

I have designed and developed an ontology to support explainability of artificial intelligence artifacts or XAI, such as machine learning models and predictions. The aim of my work is to provide a comprehensive domain-level view of this subfield of artificial intelligence. The purpose of this validation is to test its conceptual correctness, modelling consistency, ontology comprehensibility, and practical utility. An ontology is a formal representation of the concepts, properties, and relationships within a particular domain, providing a shared and machine-readable conceptual model. XAI, or Explainable Artificial Intelligence, refers to methods and approaches that make the reasoning, behaviour, or predictions of AI systems understandable to humans.

- What is an explanation?
- What concepts are essential in an explainable ML pipeline?
- Which stakeholders should be represented?
- What relationships should exist between stakeholders and explanations?

Given the relevance of my work for clinical decision-making, I have grounded my validation in a case study on cardiac autonomic neuropathy (CAN) screening in type 2 diabetes mellitus (T2DM) patients, reusing an existing but incompletely explainable pipeline. In addition to the global explanations that the authors provided, I generated local explanations before building a knowledge graph that is supported by my ontology. Cardiac autonomic neuropathy is a complication of diabetes in which damage to the autonomic nerves affects the regulation of cardiovascular functions such as heart rate and blood pressure. Type 2 diabetes mellitus is a chronic metabolic disease in which the body becomes resistant to insulin and/or does not produce enough insulin, resulting in persistently elevated blood glucose levels.

- What information would a patient need?
- What information would a nurse need?

Ontology review

Observe the conceptual models of the explanation module, the updated explanation module, and the dictionary of terms.

In my study, I analysed explanations through five interconnected dimensions: XAI stakeholders, explanation goals, explanation types, explanation techniques, and explanation quality criteria.

XAI stakeholders are defined as the human and artificial agents involved in XAI systems or projects, including those who request, receive, generate, use, or evaluate explanations. These stakeholders determine the context and requirements of an explanation.

Explanation goals refer to the objectives that an explanation is intended to achieve for a specific stakeholder. Adhikari et al. (2022) proposed eight explanation goals: Discover, Decision Support, Calibrate Trust, Justify, Audit, Improve, Contest, and Persuade.

Explanation types represent groups of explanations with similar characteristics. Chari et al. (2024) proposed fifteen explanation types: Case-Based, Contextual, Contrastive, Counterfactual, Everyday, Scientific, Simulation-Based, Statistical, Trace-Based, Data, Rationale, Safety and Performance, Impact, Fairness, and Responsibility.

Finally, explanation quality criteria describe the properties used to evaluate and assess whether an explanation fulfils its intended purpose. Nauta et al. (2023) proposed twelve explanation quality properties: Correctness, Completeness, Consistency, Continuity, Contrastivity, Covariate Complexity, Compactness, Composition, Confidence, Context, Coherence, and Controllability.

Observe the conceptual models of the stakeholder profile, explanation goal, explanation type, and explanation quality modules.

Use a Likert scale from 1 to 5 to review various criteria with relevant statements. 1 = Not at all, 2 = Slightly, 3 = Moderately, 4 = Largely, 5 = Completely. For the XAI experts, fill ratings in Table 1. For the ontology experts, fill ratings in Table 2. For the healthcare experts, fill ratings in Table 3.

These open questions are for the XAI experts:

- Which explanation concepts are missing?
- Which explanation types should be refined?
- Are any explanation quality properties incorrectly modelled?
- What would improve the ontology's usefulness for XAI practitioners?

Table 1: Statements for XAI experts given the conceptual models and ontology that need to be rated.

Criterion	**Statement**	**Rate from 1-5**
XAI domain adequacy	The ontology adequately represents the XAI field	
Completeness	The ontology covers the most important XAI concepts.	
Explanation usefulness	The ontology supports understanding AI explanations.	
Explanation relevance	The represented explanation types are meaningful and useful.	
CQ support	The ontology outputs correspond to realistic information needs.	
Stakeholder awareness	Stakeholder-specific explanation needs are represented appropriately.	
Explanation diversity	The ontology adequately distinguishes different explanation approaches.	
Explanation quality representation	Explanation quality criteria are represented appropriately.	
Explainability coverage	The ontology covers the major explainability concerns relevant to this use case.	
Explainability terminology	Explanation concepts and definitions are used consistently and correctly.	
Actionability support	The ontology captures actionable explanation knowledge.	
Future applicability	The ontology can accommodate future XAI methods and explanation types.	

Table 2: Statements for ontology experts given the conceptual models and ontology that need to be rated.

Criterion	**Statement**	**Rate from 1-5**
Conceptual correctness	The concepts are represented correctly	
Concept necessity	All introduced concepts are necessary	
Stakeholder modelling	Stakeholder roles are represented appropriately	
Type modelling	Explanation types are classified appropriately	
Ontological completeness	All important ontology elements are represented	
Foundational alignment	The OntoUML/gUFO grounding is appropriate	
Semantic consistency	Relations and constraints are logically coherent	
Constraint adequacy	Cardinalities, restrictions, and constraints are appropriate	
Modular consistency	The extension is consistent with the original XMLPO ontology	
Extensibility	The ontology can be extended without major redesign	
Interoperability	The ontology sufficiently supports integration with external ontologies and knowledge graphs	
Reuse quality	Reused ontologies are integrated appropriately	
Documentation clarity	Definitions are understandable	

Table 3: Statements for healthcare experts given the conceptual models and ontology that need to be rated.

Criterion	**Statement**	**Rate from 1-5**
Domain adequacy	Reflects CAN screening practice	
Clinical relevance	The represented concepts are meaningful in practice	
Clinical correctness	Concepts are represented in a clinically correct manner	
Terminology clarity	The terminology is understandable to healthcare professionals	
Completeness	Important clinical concepts are modelled	
Information sufficiency	Is enough information represented for the intended use case	
Stakeholder distinction	Distinguishes stakeholders appropriately	
Workflow alignment	The ontology reflects the screening workflow adequately	
Usefulness	Supports understanding of screening outcomes and explanations?	
Decision-support value	The represented information supports clinical decision-making	
Practical applicability	The ontology can be realistically adopted in practice	

These open questions are for the ontology experts:

- Which modelling decision concerns you most?
- Which concept requires further clarification?
- Which ontology modification would most improve quality?

These open questions are for the healthcare experts:

- Which clinically relevant concepts are missing?
- Are any concepts misleading or incorrectly represented?
- Which additions would most improve usefulness for clinical screening?

Use case review

Share the competency questions, SPARQL queries, and examples of visual explanations.

Questions from the patient's perspective:

1. Is the explanation relevant and understandable for you as a patient?
2. Does the explanation help you understand why the system made this prediction and what the important factors are?
3. Does the counterfactual explanation help you understand what could be different to obtain another outcome?

Questions from the nurse's perspective:

1. Does the explanation provide clinically relevant information that supports your interpretation and decision-making?
2. Do the highlighted factors and explanation appear consistent with the model's prediction and with your clinical expectations?
3. Is the confidence information meaningful and sufficient for clinical use?
4. Would you consider this explanation useful and sufficiently concise for routine clinical use and for communicating with the patient?

From the general stakeholder's perspective:

1. Which stakeholders in this use case would benefit from explanations that are not currently well supported?
2. What stakeholder-specific information or explanation needs are currently missing?
3. Does the ontology provide enough information to meaningfully customize explanations for different stakeholder groups?
4. Is there anything important about the explanation or the stakeholder needs that is currently missing?

Table 4: General statements for all participants that need to be rated.

Criterion	**Statement**	**Rate from 1-5**
Overall adequacy	The ontology adequately represents explainable machine-learning pipelines.	
Stakeholder support	The ontology adequately supports stakeholder-specific explanations.	
Conceptual completeness	The ontology captures the most important concepts of the domain.	
Practical usefulness	The ontology would be useful in practice.	
Reusability	The ontology could be reused in other explainable AI contexts.	
Overall quality	The ontology is of sufficient quality for its intended purpose.	

Overall assessment

Use a Likert scale from 1 to 5 to review various criteria with relevant statements. 1 = Not at all, 2 = Slightly, 3 = Moderately, 4 = Largely, 5 = Completely. For every participant, fill ratings in Table 4.

These open questions are for all participants:

- What is the strongest aspect of the ontology?
- What is the most important weakness of the ontology?
- What is currently missing?
- What should be improved first?
- Would you consider the ontology fit for its intended purpose?
- Are there any final recommendations you would make before publication?

Table 5: Issue list that captures the issues identified by the interviewed expert.

ID	Issue Description	Proposed Modification
I1		
I2		
I3		

Issue list

For each participant, record their issues in Table 5. The minutes and issue list were shared with the participants to check if it captured their answers.